

DEVELOPMENT OF A MODULAR IN-EAR WEARABLE DEVICE FOR DEMENTIA MONITORING: PRELIMINARY INSIGHTS



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INTRODUCTION

Globally, more than 55 million people live with dementia [1], and an even greater number of caregivers are also affected by the disease. As the condition progresses, the caregiver's role becomes increasingly complex, with physical and emotional exhaustion, isolation, and depression being common [2]. Wearable technology can offer valuable support in monitoring this health condition. This research continues the development of a wearable device called "HowMI- Home Wearables and Monitors Integrated" [3], designed to enhance end-user confidence by providing healthcare professionals with an innovative and more accurate care management tool, enabling faster responses in emergency situations

OBJECTIVES

This project continues the development of the HowMI device, an ear-worn device originally created in response to the challenges posed by the COVID-19 pandemic, with the aim of monitoring physiological signals [3]. The main objective is to restructure this ear device into a modular solution that enables the integration of the earpiece into a wristwatch, allowing the measurement of physiological signals both from the ear and the wrist.

In addition to the main objective, several complementary goals support this project:

- ✓ **Modularity and versatility:** The hardware architecture and product design enable the module to be used both in the ear and on the wrist.
- ✓ **Sensor accuracy and reliability:** Integration of sensors capable of providing readings with clinical precision and accuracy.
- ✓ **Ergonomics and comfort:** A user-centered design approach guided anatomical studies to optimize the device's shape for ergonomic comfort and prolonged wearability.
- ✓ **Performance and energy efficiency:** Minimising energy consumption through efficient component selection and the implementation of effective power management.

CONCLUSIONS AND FUTURE WORK

From the tests carried out, it was possible to identify some limitations related to data acquisition by the sensors. The temperature readings showed inconsistent measurements, which may lead to errors depending on the positioning of the device in the ear. Regarding the measured values of blood oxygen saturation and heart rate, it was observed that any movement of the device significantly affects the readings, resulting in inaccurate data. Parallel effort included the development of a new prototype to support controlled testing of the integrated sensors.

In the next steps a new anatomical study of the human ear will also be undertaken using 3D reconstructions from CT scan data, the aim is providing precise measurements to inform the ergonomic redesign of the device, ensuring greater comfort and stability, particularly for long-term use by individuals with dementia. New sensors based on different technologies will be selected, with the aim of assessing potential improvements, identifying the advantages and disadvantages of each sensing technology, and determining more suitable positions for data acquisition.

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METHODOLOGY

The module currently under development is part of a signal acquisition system designed to address requirements across physical, emotional, and cognitive domains [4]. Its development follows an iterative, user-centered design approach focused on refining the system to best meet these needs. At this early stage, the focus is on the intra-auricular device, HowMi, which monitors physiological parameters such as heart rate, blood oxygen saturation, and body temperature. The methodology incorporates anatomical and ergonomic studies to inform the device's physical form, ensuring it is comfortable and suitable for continuous wear. Additionally, rapid prototyping and usability testing are employed to refine both the form factor and user interaction, alongside ongoing hardware testing and refinement, the project stages are illustrated in Figure 1. In more advanced stages, the system will be tested in collaboration with a partner organization within private homes and residential care facilities. This will support non-invasive, continuous monitoring and facilitate real-time communication with care providers. The project aligns with the United Nations Sustainable Development Goals (SDGs), particularly SDG 3 - Good Health and Well-Being, SDG 9 - Industry, Innovation and Infrastructure, and SDG 10 - Reduced Inequalities, as illustrated in Figure 2 [5]. By fostering inclusive, technology-driven healthcare solutions, it contributes to early detection, ongoing monitoring, and equitable access to care for individuals living with dementia.

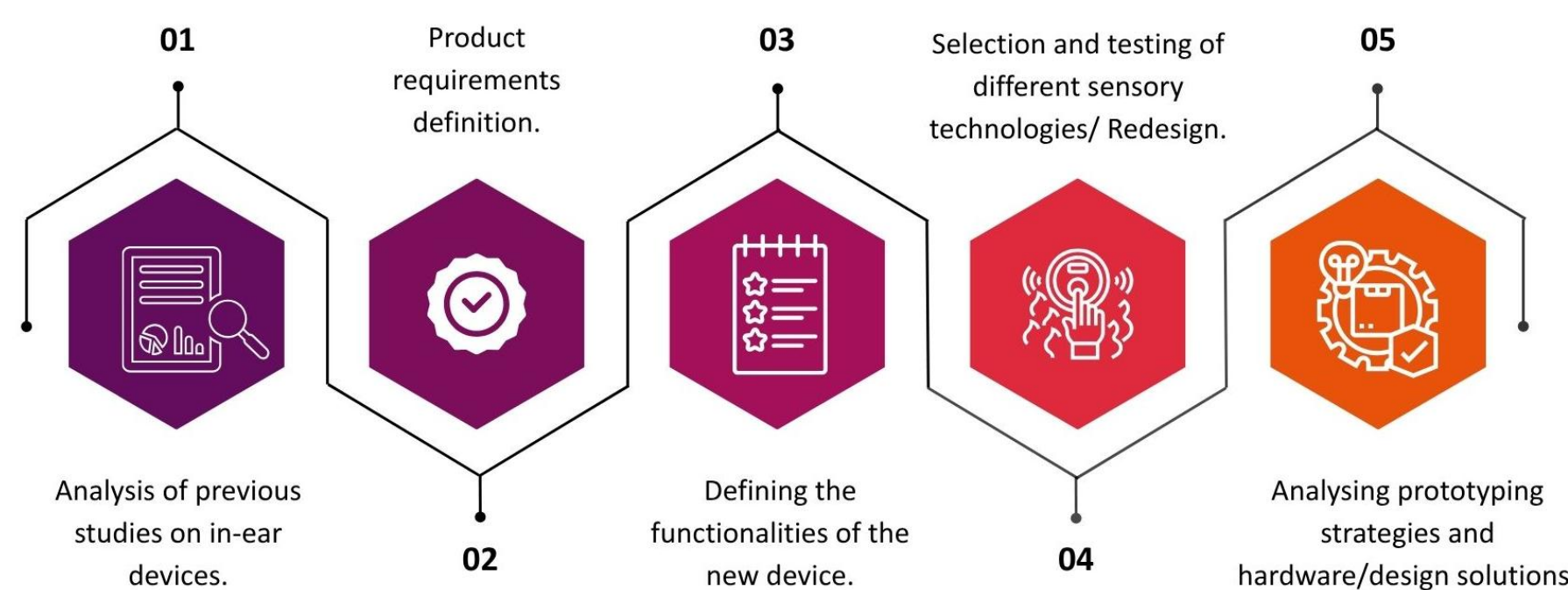


Figure 1. Overview of Key Project Development Stages.



Figure 2. Targeted Sustainable Development Goals – Adapted from [5].

RESULTS

One of the first steps in this project was to validate the HowMi. To do this, the previously designed in-ear device was developed and optimised, with particular emphasis on the hardware. The aim was to reduce the physical dimensions of the device and resolve any potential limitations without compromising the accuracy of the measurements of essential biometric data, such as oxygen saturation and body temperature. At this stage, a state-of-the-art review was also carried out both for design and hardware, to identify the most recent sensor technologies applied in wearable devices, with a particular focus on in-ear solutions, and design ergonomics trends. The redesign process is currently underway, with exploratory testing of the hardware in progress to ensure its performance before integration into the new device design. An overview of the anticipated outcomes is presented in Figure 3.

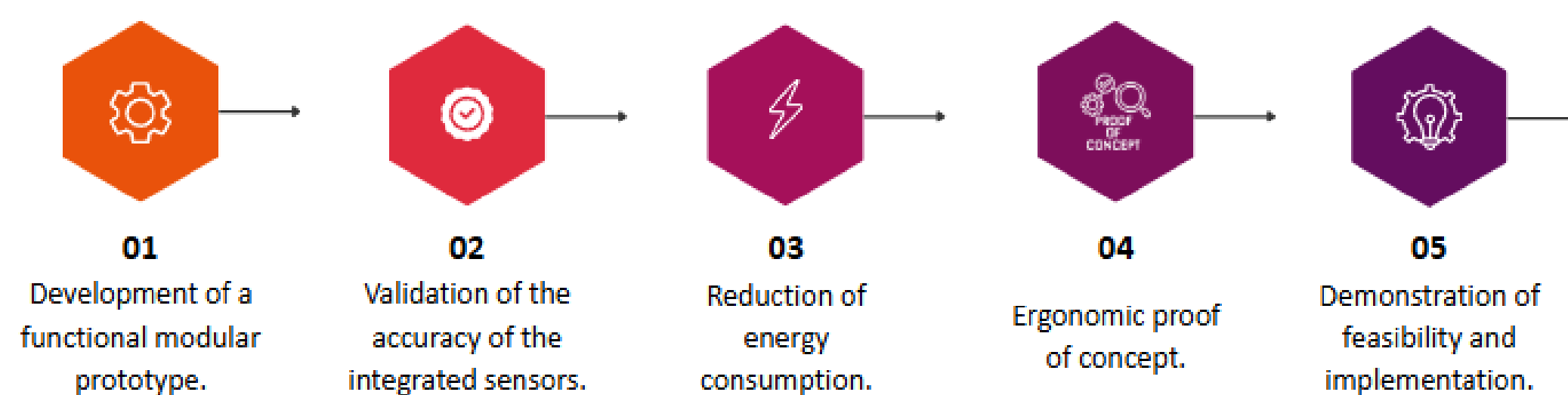


Figure 3. Anticipated Outcomes of the Project.

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